□ README

□ Project Title:

Spam Email Detection using Logistic Regression

□ Submitted By:

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□ Objective:

To build a machine learning model that automatically classifies emails as **SPAM** or **NOT SPAM** using **Logistic Regression** and **NLP techniques**.

□ Dataset Source:

SMS Spam Collection Dataset – Kaggle
Link: https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset

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□ Tools & Libraries Used:

- Python 3.x
- pandas
- numpy
- scikit-learn
- nltk
- matplotlib
- seaborn
- Flask (for web deployment)

□ Model Overview:

- **Algorithm:** Logistic Regression
- **Feature Extraction:** TF-IDF Vectorization (1–2 grams, max 3000 features)
- **Train/Test Split:** 70% training – 30% testing
- **Saved Model Files:**
  - `spam_model.pkl`
  - `tfidf_vectorizer.pkl`

□ Evaluation Summary:

Metric	Score
Accuracy	95.93%
Precision	98.14%
Recall	70.98%
F1 Score	82.38%
ROC AUC	85.38%

□ Deployment Instructions (Flask App):

1. Open Terminal or VS Code
2. Run:
3. `pip install -r requirements.txt`
4. `python app.py`
5. Open your browser and go to:
   `http://127.0.0.1:5000/`
6. Enter an email and get the prediction result instantly

□ Project Files Included:

- `Spam_Detector_Project.ipynb` – Jupyter notebook with full ML pipeline
- `Spam_Detector_Report.pdf` – Project summary + screenshots
- `README.pdf` – Overview and instructions
- `app.py` – Flask backend
- `templates/index.html` – Web interface
- `static/style.css` – Web styling
- `spam_model.pkl` – Trained model
- `tfidf_vectorizer.pkl` – Vectorizer

□ Sample Inputs to Test:

SPAM Example:

Congratulations! You've won a free iPhone. Click here to claim your prize.

NOT SPAM Example:

Hey, just checking if you'll join the meeting at 2 PM today.

Final Note:

This mini-project successfully demonstrates a complete ML pipeline from data preprocessing to model deployment. The model is lightweight, accurate, and ready to be used in real-world scenarios.